ARtPUT: Autonomous Rover to Pick Up Trash

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Abstract: Litter in U.S. public spaces generates over 51 billion pieces of debris and $11.5 billion in annual cleanup costs (Keep America Beautiful, 2021); existing autonomous solutions like OATCR rely on expensive vision hardware that limits accessibility (Kulshreshtha et al., 2021). ARtPUT (Autonomous Rover to Pick Up Trash) tests whether a rover using only low-cost Arduino components and infrared (IR) proximity sensing can autonomously detect and collect ≥70% of common park litter. An Arduino Uno R3, DFRobot Devastator chassis, Sharp IR sensor, and FIT0300 servo gripper drive forward until IR readings cross a threshold of 400, halting the motors and closing the gripper. Sixty trials were run across five litter categories at 50 cm on indoor carpet, scored Y/N on detection, stopping distance, and grasp. ARtPUT achieved 80% collection success (40/50) across three primary categories — crushed bottle (85%), paper ball (80%), plastic bag (75%) — with 96% detection, while flat paper discs and mandarin peels failed (0/5 each) due to insufficient vertical profile. Results support H1 and indicate that object geometry, not detection sophistication, is the dominant constraint for mechanical collection; future work will add camera-based vision, a wider compliant gripper, and weatherproofing.

Introduction & Background

Litter cleanup costs U.S. cities $11.5 billion annually, with over 51 billion pieces generated each year[1]. Autonomous robots have reduced warehouse labor up to 80%[2], but outdoor deployment remains unsolved due to uneven terrain and irregular object geometry.

Prior work (OATCR)[3] uses costly vision systems, limiting accessibility. ARtPUT tests whether autonomous detection and grasping is achievable using low-cost, off-the-shelf Arduino components and IR sensing.

Objective & Hypothesis

Objective: Can a low-cost Arduino rover autonomously detect and collect common park litter?
H1: Collection ≥70% of visible items in range.
H0: Success ≤50%.

Variables & Trials

IV: Litter type (5 categories)
DV: Detection, distance, grasp (Y/N)
Controls: 50cm start, IR=400, carpet, lighting
Constants: Arduino Uno, Sharp IR, FIT0300
n = 60: paper ball (10), paper disc (5), bottle (20), bag (20), peel (5)

Technical Needs & Limitations

Needs: Arduino IDE, IR calibration, stable power.
Limits: IR misses flat objects; jaw span ~6cm; slight drift; Pi failed → Arduino pivot; indoor-only.

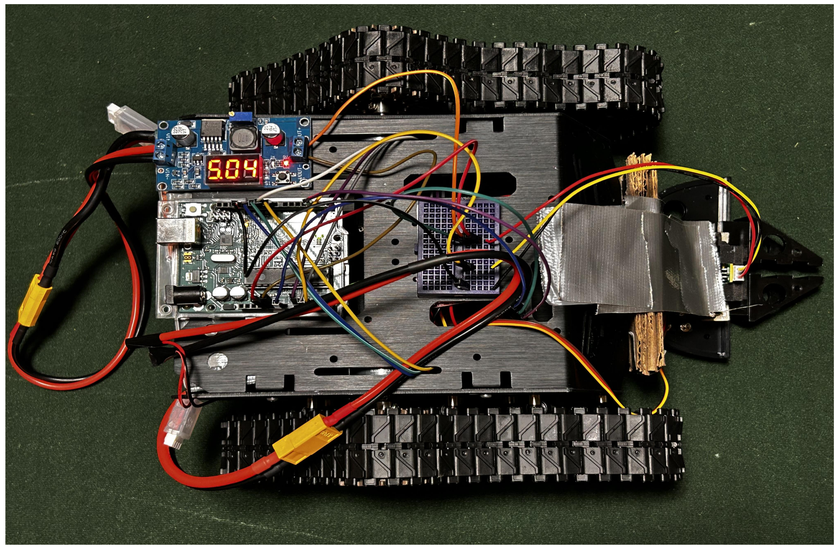

Figure 1. Top view of completed ARtPUT showing chassis with mounted Arduino, motor driver, and wiring harness.

Materials & Methods

System. Rover drives forward autonomously, polling the Sharp IR sensor. When reading exceeds 400, Arduino halts motors and closes the FIT0300 gripper.

Hardware. Arduino Uno R3 · L298N driver · DFRobot Devastator chassis · Sharp IR analog sensor · FIT0300 gripper · 11.1V batteries.

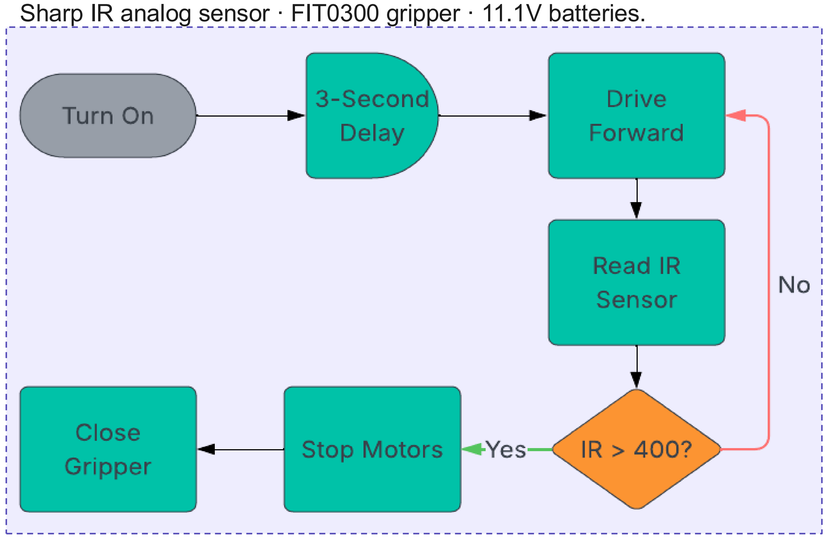

Figure 2. System flowchart of autonomous detect-stop-grasp sequence.

Protocol. Each trial: one item placed 50cm ahead on carpet. Rover powered on, observed autonomously. Scored Y/N on detection, distance, grasp.

Tuning. IR threshold tuned 350→400 in pilot trials.

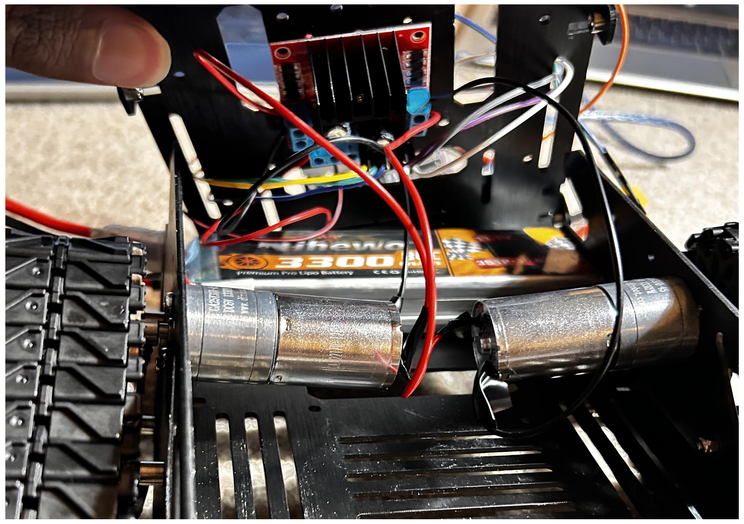

Figure 3. Interior of ARtPUT's chassis, which contains L298N motor driver, motors, and batteries.

Results & Analysis

ARtPUT achieved 80% collection success (40/50) across three primary categories, exceeding the 70% hypothesis. Detection rate: 96%.

By item. Crushed bottle performed best at 85% success (17/20), followed by paper ball at 80% (8/10) and plastic bag at 75% (15/20). Detection rate for these three categories averaged 96% (48/50), indicating the IR system reliably detected objects with sufficient vertical profile; grasp failures were driven primarily by object displacement before jaw closure.

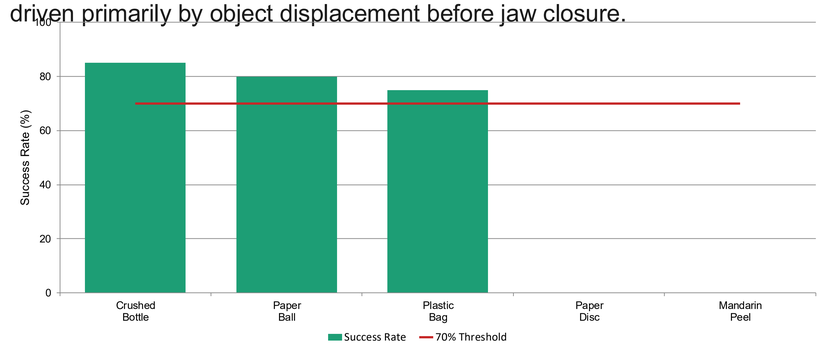

Figure 4. Collection success rate by litter type (n=60).

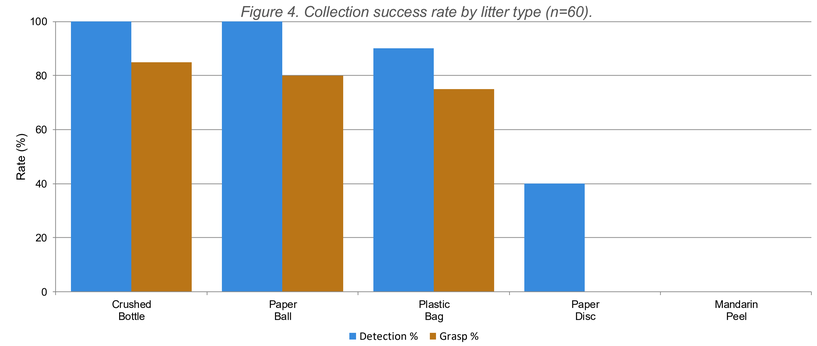

Figure 6. Detection rate vs grasp success by item category.

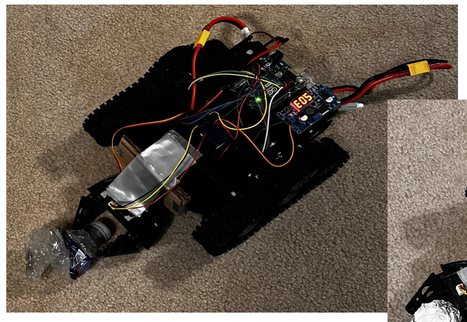

Figure 5. Successful water bottle trial.

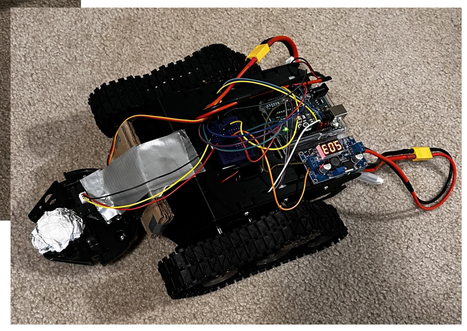

Figure 7. Successful paper ball trial.

Discussion & Conclusions

Hypothesis supported. ARtPUT demonstrates low-cost autonomous litter collection is feasible with inexpensive hardware, achieving 80% success across three primary categories. H1 supported.

Broader framework. Object geometry (vertical profile, width vs jaw span) is more critical than detection algorithm sophistication for mechanical collection tasks[4].

Future work. (1) camera-based CV for flat objects; (2) wider/compliant gripper; (3) GPS path planning for outdoor; (4) weatherproofing; (5) expand item categories.

Literature Cited. [1] Keep America Beautiful. 2021. 2020 National Litter Study. KAB Research Series. [2] Avidbots. 2025. Commercial Cleaning Robot Deployment Data. Avidbots Industry Report. [3] Kulshreshtha, M., et al. 2021. OATCR: Outdoor Autonomous Trash-Collecting Robot Design Using YOLOv4-Tiny. Electronics 10(18): 2292. [4] Memmesheimer, R., et al. 2019. Autonomous object manipulation in unstructured environments. Journal of Field Robotics 36(3): 481–502.

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